The Fold:
Scale, twist, collapse — and what is left at the centre

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0 Summary

Scaling about a datum is a one-parameter family $g_k(P) = P_0 + k(P - P_0)$. The family is a group for $k \neq 0$; $k = 0$ is the boundary where it degenerates into a point. A real k changes size, an imaginary one twists, and a fold is the reflection whose composition produces the twist. Everything that follows is the consequence of one determinant:

$$\det(g_k) = k^n, \qquad n = \text{dimension}. \tag{1}$$

Heights scale like k, areas like k^2, volumes like k^3. That single fact separates every quantity in the construction.

1 The scale factor and the datum

For a datum point P_0 and a point P on the object:

$$P' = P_0 + k(P - P_0). \tag{2}$$

k	behaviour
$k > 1$	expansion away from the datum
$0 < k < 1$	contraction toward the datum
$k = 0$	collapse: every point maps to P_0
$k < 0$	scaling composed with reflection through P_0

The map is a **homothety**. For $k \neq 0$ the family is a group isomorphic to $\mathbb{R}^*$; g_0 is the constant map with rank 0 and is not a member. This distinction matters later: zero is not a small scaling, it is the end of the family.

Area and height

$$\text{height} \quad h' = k\,h, \tag{3}$$

$$\text{area} \quad A' = k^2\,A, \tag{4}$$

$$\text{volume} \quad V' = |k|^3\,V, \tag{5}$$

so that

$$\frac{A'}{A} = \left(\frac{h'}{h}\right)^2. \tag{6}$$

Height and area cannot share a law. A sign survives in the height and dies in the area, because area is a determinant and the determinant of kI is k^2. At $k = -1$ the height reverses and the area is unchanged.

2 Two regimes: the sphere and the cone

Where the datum sits decides the geometry, and there are exactly two cases.

Datum inside the body. Every orbit of the scaling is a diameter. The level sets are spheres:

$$g_k(S_r) = S_{|k|r} \qquad (k \neq 0). \tag{7}$$

The tangent cone at an interior point is the whole space $\mathbb{R}^3$ — no proper cone exists. The flow is purely radial and the collapse is isotropic. The bulk-to-boundary ratio vanishes:

$$\frac{V}{A} = \frac{r}{3} \to 0. \tag{8}$$

Datum on the boundary. The orbits are rays and their union is a proper cone with apex P_0. This is the case that produces the cone of rays, and the case a horizon belongs to.

Datum outside. Same cone, seen from further away: the body subtends a cone and casts a shadow.

The three named choices of datum are three positions of the same apex:

datum	apex	tangent cone	behaviour
centroid	interior	$\mathbb{R}^3$	uniform scaling in place
vertex / base	boundary	proper cone	apex locked, body grows away
origin	fixed frame point, interior or exterior	$\mathbb{R}^3$ or proper cone	scaling also translates the body when it is off-origin

3 The twist

The operator

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I \tag{9}$$

is the infinitesimal generator of rotation, and

$$e^{\theta J} = \cos\theta\, I + \sin\theta\, J = R(\theta). \tag{10}$$

A general scale-with-twist is therefore the spiral similarity

$$k = \rho e^{i\theta}, \qquad P' = P_0 + \rho\,(\cos\theta\, I + \sin\theta\, J)\,(P - P_0). \tag{11}$$

J itself is the quarter turn. The four cardinal values are the fourth roots of unity and form a cycle:

$$1 \to i \to -1 \to -i \to 1. \tag{12}$$

Two quarter turns equal one half turn: $i^2 = -1$. The twist squared is the flip.

Which twists are observable

The twists that leave a body unchanged *are* its rotation symmetry group. Written as a quotient, the body only responds to

$$U(1)/\operatorname{Sym}(B). \tag{13}$$

body	symmetry	observable twist
sphere	S^1 — all θ	none: $i = 1$ exactly
cube	$C_4 = \{1, i, -1, -i\}$	only between the cardinal turns
chiral body	$\{1\}$	every angle

Measured on the rendered model (silhouette change under θ):

θ	0°	30°	45°	90°	180°	270°
sphere	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
cube	0.0%	1.4%	**12.3%**	0.0%	0.0%	0.0%

The peak at 45° is the antipode of the fundamental domain: it is the point furthest from any symmetry.

Orientation

Because $\det(kI) = k^n$, the sign of k has different meaning in different dimensions:

- **2D**: $\det(-I) = +1$ — $k = -1$ is a rotation by 180°, orientation preserved.
- **3D**: $\det(-I) = -1$ — $k = -1$ is a point reflection, orientation reversed: the genuine "inside out".

The same formula, two different geometries.

4 The fold, completed

A **fold** is a reflection in a hyperplane H:

$$\sigma_H^2 = I, \qquad \sigma_H \text{ fixes } H, \qquad \det \sigma_H = -1. \tag{14}$$

Its fixed set has dimension $n - 1$: a line in the plane, a plane in space. This is the piece the earlier account was missing — the fold does not fix the datum, it fixes a **crease**.

Two folds make a twist. Composing reflections in two hyperplanes meeting at angle α gives a rotation by 2α about their intersection:

$$\sigma_{H_2} \circ \sigma_{H_1} = R(2\alpha). \tag{15}$$

So:

- J (90°) is **two folds at** 45° — never a single fold
- $-I$ in the plane (180°) is two folds at 90°, or equivalently J^2
- J is orientation-preserving because it is the product of an **even** number of folds

Orientation is the parity of the fold count. This is why a twist and a fold cannot be confused: one has determinant $+1$, the other -1, and any odd number of folds reverses orientation.

The hinge is the intersection. The twist fixes the intersection of the two creases: a point in the plane, a line in space. The datum is that hinge.

The degeneration. Send the two creases together and their intersection collapses to a point. The fold then has no crease left, and what remains is $k = -1$: a point reflection. That

is the precise content of "the datum becomes the hinge of the fold" — the crease has shrunk into the datum, and the fold has become the point reflection.

Completion. The fold group closes when the creases have swept the full angle:

$$(\sigma_2\sigma_1)^4 = I \qquad \text{for } \alpha = 45^\circ. \tag{16}$$

Four quarter turns return the body to itself; the two creases at 45° produce the four reflection axes at multiples of 45°, and the dihedral group D_4 of order eight is generated by exactly two folds. The cycle $1 \to i \to -1 \to -i \to 1$ is that closure written multiplicatively.

The only set fixed by the entire group is the hinge: a point in the plane, a line in space.

5 The horizon

A horizon is the boundary case of section 2: the apex sits on the surface, and the surface becomes null. It is where the cone reappears.

Under $r \to kr$ the mass scales too, because $r_s = 2GM/c^2$ is a length. That family is a set of solutions, not a process: every member has the same compactness, so the horizon is present in all of them or in none. A collapse is the opposite — the mass is held fixed and only R falls. The two give different laws and must not be read off one table:

quantity	homothetic family, $M \propto k$	collapse, M fixed
horizon radius r_H	$\propto k$	$= r_s$, constant
horizon area A_H	$\propto k^2$	constant
entropy $S = A_H/4\ell_P^2$	$\propto k^2$	constant
temperature T	$\propto 1/k$	constant
mass M	$\propto k$	constant
body radius R	$\propto k$	$\propto k$
compactness r_H/R	constant	$\propto 1/k$

Only the right-hand column is a process. In the left-hand column r_H/R is a label of the member, not a stage, and the first law is exact: with $S = \alpha M^2$ and $T = \beta/M$, $T\,dS = dM$ holds identically, and $S \cdot T \propto k$ is the horizon's energy. In the right-hand column M is fixed, so both sides of $T\,dS = dM$ vanish once the horizon exists: the entropy is not accumulated during the fall, it appears with the horizon in one step.

The horizon arrives **before** zero. A body of radius R reaches its own Schwarzschild radius at

$$k^* = \frac{r_s}{R} = \frac{2GM}{c^2 R}. \tag{17}$$

body	k^*
Earth	1.4×10^{-9}
Sun	4.3×10^{-6}
70 kg water sphere	4×10^{-25}

Below k^* the collapse has left the observable chart. The point remains in the mathematics and leaves the physics.

Because $V \propto k^3$ while the horizon area is constant, $V/A_H \propto k^3$ in the collapse, and at the moment of formation it is $r_s/3$; in the homothetic family both scale, so the ratio falls only as k. Either way $V/A \to 0$. That is the quantitative content of the holographic statement: information capacity tracks the area, and the area outlives the volume.

6 Zero as the point

$k = 0$ is not a scaling. It is:

- **outside the group** — rank 0, not invertible
- **the compactification of every orbit** — each ray $(P_0, P]$ is open, and $k = 0$ closes them all at once
- **without geometry** — no tangent space, no normal; curvature $1/r$ is undefined, not infinite

The dimensional ladder is therefore not a path through intermediate objects:

$$V \propto k^3, \qquad A \propto k^2, \qquad h \propto k, \qquad \text{point} = k^0. \tag{18}$$

These are four measurements of the same family, not four consecutive states. At $k = 0$ the rank drops from n to 0 in one step; the body never passes through a sheet or a line.

7 What an observer can see

A point has no aperture: it subtends zero solid angle and collects zero flux. What it *can* measure is fixed by its dimension:

observer dimension	observable
0D	scalar — curvature, potential, temperature
1D	derivative, direction
2D	flux (needs an oriented area)
3D	density, integral

A point sees **values, not flows**. It has no tangent space, so it cannot host a vector. Two complementary views of the same sphere:

observer	$\Omega/4\pi$	behaviour
outside	5.06% at $k = 1$, falls as k^2	body shrinks to a point
at the centre	100% for every $k > 0$	sky fixed, distance shrinks
on the surface	2π half-space	edge-on; the horizon case

Measured exterior view:

k	$\Omega/4\pi$	ratio
1.00	5.060%	1.000×
0.50	1.216%	0.240×
0.25	0.301%	0.060×
0	0.000%	—

The interior view is constant because a homothety is **conformal**: it preserves angles, hence solid angle, exactly. Scaling preserves what you see, but not how far away it is. Let the interior sphere recede ($k \to \infty$) and the curvature $1/r \to 0$ while $\Omega = 4\pi$ stays fixed — the boundary flattens into the celestial plane at infinity, still carrying everything the bulk can send.

At $k = 0$ all three views coincide: the point, the shell, and the observer are the same place.

8 Devices that instantiate each step

idea	device	what it demonstrates
scale invariance	pinhole camera	image scales as k, area as k^2
interior collapse	inertial confinement, sonoluminescence	compression $1/k^3$
radial flow	integrating sphere	$\Omega = 4\pi$ independent of radius
twist J	quarter-wave plate	Jones matrix $\mathrm{diag}(1, i)$
two folds make a twist	two mirrors at 45°	rotation by twice the mirror angle
symmetry quotient	square waveguide	invisible at 0/90/180/270, maximal at 45°
symmetry breaking	chiral medium (sugar solution)	every angle observable
directional twist	Faraday rotator	non-reciprocal 90°
horizon	acoustic / optical analogue horizon	$r \propto k$, $T \propto 1/k$
conformal boundary	AdS/CFT, celestial sphere	physics at infinity, Ω fixed
0D scalar probe	thermocouple, curvature scalar	value without direction

9 The one sentence

Scaling preserves angles and curvature-free structure, changes lengths and curvatures, is generated by folds in pairs, and terminates in a point that has no geometry and can only be seen as a direction.

Companion models: `j-plane.html` *(the plane, area and height),* `j-cone-3d.html` *(the external cone),* `j-collapse-3d.html` *(sphere, cube, horizon readouts, observers).*